

Fernando Sepulveda

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EDUCATION

The University of Texas at El Paso (UTEP)

- Master of Science in Computer Science
- Bachelor of Science in Computer Science

GPA: 3.5 | Expected: 05/2027

GPA: 3.03 | Graduated: 12/2024

Honors & Achievements

UNCF Oracle Scholar 2023 | HITEC Scholar, 2022& 23& 24 | MSIPP Scholar, 2022 | Deans List 2023 & 2025

PROFESSIONAL EXPERIENCE

U.S. Army Engineer Research & Development Center (ERDC)

Alexandria, VA

AI Research Intern – Geospatial Research Laboratory

05/2026 – Present

- Engineered a zero-shot geospatial segmentation workflow using PyTorch and Ultralytics FastSAM, integrating K-Means clustering to dynamically classify materials and reduce false-positive detections
- Designed and evaluated 5 distinct computer vision architectures to extract and validate structural terrain bottlenecks from high-resolution RGB satellite imagery
- Deployed containerized autonomous navigation systems using Docker, ROS, and real-time LiDAR sensor arrays to dynamically map and route through complex physical environments

Kansas City National Security Campus - Honeywell

Kansas City, KS

Student Engineering Intern III – Data Scientist (Q level Clearance)

06/2024 – 08/2024

- Spearheaded a black belt automation initiative driving projected annual cost savings of \$250,000 by optimizing time efficiency across factory operations leveraging Dataiku, Hadoop, and Python programming
- Engineered a robust, security-first semantic search system employing Large Language Models (BERT, SBERT), ensuring highly secure data management and retrieval.

Oracle Cloud Infrastructure (OCI) - Oracle

Austin, TX

Software Engineer Intern

05/2023 – 08/2023

- Engineered a Java-based automation script to manage and dynamically organize cloud hypervisors by region based on maximum pool capacity and shape requirements.
- Built a reusable API façade class to process RESTful calls between front-end interfaces and backend computation, enabling efficient data segmentation and retrieval.

Smart Autonomous Mobility Systems (SAMS) Lab - UTEP

El Paso, TX

Undergraduate Research Student

08/2023 – 12/2024

- Published research evaluating the impact of strategic, false data injection cyberattacks on autonomous vehicle platoons, analyzing safety (TTC), mobility (ASV), and energy consumption metrics
- Developed Python-based simulation frameworks to analyze mixed traffic flow dynamics, implementing the Intelligent Driver Model (IDM) for human-driven vehicles and the OVRV model for autonomous vehicles

PROJECTS/ LEADERSHIP

Mike Loya Center for Innovation and Commerce

El Paso, TX

AI Graduate Research Fellow

01/2026 – Present

- Designed a multi-phase AI commercialization pipeline, evaluating the technical feasibility of hackathon projects to identify 6 major startup candidates for further investment
- Provided targeted 1-on-1 mentorship to the top 3 AI ventures, overcoming algorithmic hurdles to build concept demonstration prototypes and rigorous competitive market analyses

Machine Learning Research Project (UTEP)

El Paso, TX

AI Graduate Researcher

03/2024 – Present

- Implemented a TensorFlow/Keras-based CNN architecture for facial recognition, mitigating severe overfitting by integrating Batch Normalization and Dropout layers to achieve 100% test accuracy
- Engineered an automated data pipeline to capture and preprocess 10,000+ image samples, applying Principal Component Analysis (PCA) and Grad-CAM for feature extraction and cluster evaluation

SKILLS

- **Programming:** Python, Java, Modern C++ (17/20), SQL, TypeScript, Shell Scripting/ Bash, ROS2
- **Machine Learning & AI:** Transformer-based LLMs, Computer Vision (CNNs), TensorFlow, PyTorch, scikit-learn, Pandas & NumPy, SBERT, Sensor Fusion (LiDAR/Camera), Diffusion Models
- **Cloud & Databases:** Oracle Cloud Infrastructure (OCI), Amazon Web Services (AWS), Database Systems (MySQL, Neo4j, PostgreSQL, MongoDB)
- **Languages:** English (C2), Spanish (C2), French (C1), Portuguese (A1)